LAW AND SECURITY IN OUTER SPACE FROM THE VIEWPOINT OF PRIVATE INDUSTRY +

*By Roger K. Hoover**

I. INTRODUCTION

Much has been written about the effect of the use of outer space on national and international security. Also, there is much written on the principle of reserving outer space for peaceful uses only and the effect of this principle on self-defense and, in turn, on the ability of nations to maintain their security. These are important and complex issues to be addressed and, hopefully, resolved within the context of international law and space law. In this paper I would like to address the issues of law and security in outer space, not from the point of view of the world, of national alliances, or of nations, but from the point of view of private industry. I will review what "security" and some of the elements thereof are to private industry, how these relate to private industry engaged in space activities, the extent to which they are covered by existing space law, the effect of such coverage on private industry and some areas which still need to be addressed by space law to provide security for private industry in outer space.

II. PRIVATE INDUSTRY SECURITY

What does security mean to private industry? Webster defines "security" as "the quality or state of being secure; freedom from danger; safety; freedom from fear or anxiety; freedom from want or deprivation." This definition fits nicely into the concept of security for private industry. The desires of private industry for security translate into a desire for freedom from danger, fear, anxiety and deprivation relating to its right to conduct business, its equipment, its employees, its technology, and its profits.

Once private industry has taken necessary actions internally, it relies on the legal regime in which it is operating for additional assurances of security. The legal regime to support the security of private industry must provide for the authority of private industry to operate in the geographical area and in the business area in which it is interested. It should provide for protection against interference by others in the private industry's legitimate business and protection from harm or damage by others to the industry's technology, equipment, employees and general right to operate. With regard to non-space activities, from the very nature of the existence of private industry to varying degrees around the world, we can conclude that the legal regime as encompassed in local, national, and international law provides to private industry a sufficient degree of security to permit it to continue and even to thrive.

---

*Division Counsel, Lockheed Missiles & Space Co., Inc., Austin Division.

+ The views expressed in this article are those of the author and not necessarily those of Lockheed Missiles and Space Co., Inc.

III. SECURITY FOR PRIVATE INDUSTRY IN SPACE ACTIVITIES

Any private industry which is considering activities in outer space will seek in those activities the same degree of security which it enjoys in its earthbound activities. The degree to which the elements of security are provided for private industrial activities in outer space will have an important effect on the degree to which private industry will participate in outer space activities. Such security will be dependent upon the legal regime which governs outer space activities. Thus, to analyze the question of security of private industry in outer space we need to review the existing legal status of outer space. In doing this, we will look principally at the four major existing space treaties, which I will refer to as the "Outer Space Treaty,"[1] the "Rescue and Return Treaty,"[2] the "Liability Treaty"[3] and the "Registration Treaty."[4]

IV. EFFECT OF SPACE LAW ON SECURITY OF PRIVATE INDUSTRY ACTIVITIES IN OUTER SPACE

A. *Right to Participate*

The first issue of "security" for private industry involves the question of whether or not it has a right to participate in a particular area of business. To the extent that the right to participate is restricted or doubtful, private industry will feel insecure about its participation.

Does private industry have a right to participate in outer space activities? I believe that under existing space law the answer in general is "yes".

Early in the negotiations of the Outer Space Treaty, it was recommended by some nations that participation in outer space activities should be limited to nations or to the "states". Private industry, it was suggested, should not be permitted to participate. Those suggestions were not adopted and the Outer Space Treaty does not restrict participation to governments.[5] In fact, Articles VI and IX of that treaty make specific

---

[1]Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies (hereinafter "Outer Space Treaty"), Jan. 27, 1967 [1967] 18 U.S.T. 2410, T.I.A.S. 6347, 610 U.N.T.S. 205 (effective Oct. 10, 1967).

[2]Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched Into Outer Space (hereinafter "Rescue and Return Agreement"), April 22, 1968 [1969] 19 U.S.T. 7570, T.I.A.S. 6599, 672 U.N.T.S. 119 (effective Dec. 3, 1968).

[3]Convention on International Liability for Damage Caused by Space Objects (hereinafter "Liability Convention"), March 29, 1972 [1973] 24 U.S.T. 2389, T.I.A.S. 7762 (effective Oct. 9, 1973).

[4]Convention on Registration of Objects Launched Into Outer Space (hereinafter "Registration Convention"), Jan. 14, 1975 [1976], T.I.A.S. 8480 (effective Sept. 15, 1976).

[5]"Outer Space Treaty," *supra* note 1, art. VI.

references to "non-governmental entities"[6] and to the activities of a state "or its nationals in outer space."[7] Article VI of the Outer Space Treaty does require that "activities of non-governmental entities in outer space, including the moon and other celestial bodies, shall require authorization and continuing supervision by the appropriate State Party to the Treaty."[8]

In addition, Article VII of the Outer Space Treaty makes a State Party "from whose territory or facility an object is launched" liable for damage by such object.[9] This is reinforced by the Registration Treaty which requires all space objects to be registered to a nation[10] and, thus, would require any private industry space objects to be registered to a nation. The Liability Treaty further reinforces the liability of a "launching state"[11] for damage caused by a space object of the launching state.[12]

It can reasonably be expected that any nation, being so subjected to liability, will control and restrict the activities of private industry so as not only to reduce the risk of damage which may be caused by a launch involving private industry of a space object from the nation's territory, but also to pass on to such private industry the liability for such damages. In such a situation, the level of security enjoyed by the private industry will be greatly influenced by the technical risks involved in the contemplated space activity as well as the ability of the private industry to cover such risks by its own financial responsibility, by insurance, or through indemnification from some other party or a combination thereof.

> States Parties to this Agreement shall bear international responsibility for national activities on the moon whether such activities are carried on by governmental agencies or by non-governmental entities, and for assuring that national activities are in conformity with the provisions set forth in the present Agreement. States Parties shall insure that non-governmental entities under their jurisdiction shall engage in activities on the moon only under the authority and continuing supervision of the appropriate State Party.[14]

---

[6]*Id.*

[7]*Id.*, art. IX.

[8]*Id.*, art. VI.

[9]*Id.*, art. VII.

[10]"Registration Convention," *supra* note 4, art. II, para. 1.

[11]The term "launching state" was defined by the Convention as:
(1) "A state which launches or procures the launching of a space object" and
(2) "A state from whose territory or facility a space object is launched."
"Liability Convention," *supra* note 3, art. 1, §c.

[12]*Id.*, *supra* note 3, art. II.

[13]Draft Agreement Governing the Activities of States on the Moon and Other Celestial Bodies (hereinafter "Moon Treaty"), U.N. GAOR, 34th Sess., Supp. H. No. 20 (Doc. A/34/20).

[14]*Id.*, art. XIV, para. 1.

Thus, private industry is permitted to participate in outer space activities but only as authorized and supervised by a nation and as controlled or regulated by the nation with regard to the risk of liability for damages faced by the nation as a result of the private industry involvement.

B. *Effect of National Claims of Ownership to Outer Space*

Private industry must determine where in outer space to conduct its activities. This decision may very well be affected by claims of ownership to outer space by individual nations. For example, a private industry intending to conduct outer space activities in the geostationary orbit might not feel very secure about doing so if it were aware of the claims by equatorial countries of ownership of that orbit. Private industry, being aware that in order to get to and remain in an outer space position its space object must pass through or remain in outer space over the area which is within the borders of another nation or nations, might likewise feel insecure in knowing that various nations from time-to-time have claimed ownership and sovereignty over the outer space above their sovereign territories. The degree of security or insecurity would be affected by the extent to which space law supports or rejects such claims.

Claims of ownership of outer space have generally not been recognized by international or space law. The Outer Space Treaty provides that "[o]uter space, including the Moon and other celestial bodies, shall be free for exploration and use by all states without discrimination of any kind, on a basis of equality and in accordance with international law, and there shall be free access to all areas of celestial bodies."[15] Similarly, the proposed Moon Treaty would provide that "[t]he Moon is not subject to national appropriation by any claim of sovereignty, by means of use of occupation, or by any other means."[16] Paragraph 3, of Article XI would provide that "[n]either the surface nor the subsurface of the Moon, nor any part thereof or natural resources in place, shall become property of any state, international intergovernmental or non-governmental organization, national organization or non-governmental entity or of any natural person."[17]

Thus, under the Outer Space Treaty and under the proposed Moon Treaty, private industry need not feel a great deal of insecurity arising out of national claims of ownership to or sovereignity over areas of outer space or celestial bodies.

C. *Harm and Damage to Persons or Property in Transit and in Outer Space*

Private industry would be concerned about the security of its property and employees (if any) while making the commute to and while stationed at an outer space location.

Because of the advanced technological state of outer space activities, there can always be technical risks which affect the security of property and persons in outer space

---

[15] "Outer Space Treaty," *supra* note 1, art. 1.

[16] "Moon Treaty," *supra* note 13, art. XI, para. 2.

[17] *Id.*, para. 3.

activity. These would have to be addressed internally in private industry through technical safety and reliability measures. Further, national or international law may impose safety standards or restrictions on the private industrial activities which would have to be met.

If harm or damage arises from collision with a traceable space object, private industry is afforded some level of security by the existing treaties. The Rescue and Return Treaty would require other nations to provide assistance to a damaged spacecraft[18] and its personnel[19] and to return a downed spacecraft[20] and its personnel.[21] The Registration Treaty would hopefully permit identification of the responsible nation.[22] The Liability Treaty would provide a method of obtaining recovery for such damage.[23] However, the ability of the private industry to recover damages is dependent upon the ability of nations to resolve the claims or disputes between them and, failing such resolution, their willingness to accept the determination of a claim commission under the Liability Treaty.

To the extent that damage is caused not by a space object, but by another person, the ability of the private industry to obtain redress is unclear. The Outer Space Treaty forbids interference, but does not provide a clear remedy if such interference does occur. The Liability Treaty provides for redress of damages caused by a space object,[24] but does not clearly provide such redress for damages caused by persons to the property or employees of another in outer space. Private industry would have to rely on the ability of nations to consult and resolve the issues or, if available, on the application of some international law. In this area, private industry is not provided a clear degree of security.

There is a security, or safety, risk which needs to be addressed by space law. This is the risk of collision created by abandoned orbiting manmade space objects which are no longer controlled or controllable. As space activities increase, more space objects are launched, more are abandoned in space, and the danger of collision with abandoned objects increases. Although the Registration Treaty requires that all space objects be registered to a nation[25] and the Liability Treaty places liability for damages caused by space objects on the launching nation,[26] it may be difficult, or even impossible, to determine the nation responsible for a specific abandoned, manmade space object, or portion thereof, after many years in orbit. The lack of control over this area increases the

---

[18] "Rescue and Return Agreement," *supra* note 2, art. V, para. 2.

[19] *Id.*, art. II, III.

[20] *Id.*, art. V, para. 3.

[21] *Id.*, art. IV.

[22] *See*, "Registration Convention," *supra* note 4, art. VI.

[23] "Liability Convention," *supra* note 3, art. VIII through XXIII.

[24] "Liability Convention," *supra* note 3, art. II.

[25] "Registration Convention," *supra* note 4, art. II, para. 1.

[26] "Liability Convention," *supra* note 3, art. II.

exposure to financial loss for private industry and its security is thereby decreased. Space law should be expanded to require the responsible party to dispose of a space object when it is no longer useful in some manner that will render it harmless to other orbiting space objects.

D. *Ability to Operate Without Interference*

Private industry would want to be assured that, once having established its operations in some outer space location, it could conduct those operations without interference from others. The Outer Space Treaty on this issue provides that parties shall carry on activities in outer space "in accordance with international law, including the Charter of the United Nations, in the interest of maintaining international peace and security and promoting international cooperation and understanding."[27] Article IX provides that states shall "conduct all their activities in outer space. . .with due regard to the corresponding interests of all other states. . . ." And, also in Article IX, a state which has reason to believe that an activity or experiment planned by it or its nationals in outer space would cause potentially harmful interference with activities of other State Parties shall "undertake appropriate international consultations before proceeding with any such activity or experiment" and any party which believes that the activities of another party in outer space would cause potentially harmful interference with other activities in outer space may request consultation concerning such activity.

These provisions are not particularly strong security for private industry against interference by others in space except to the extent that such interference may otherwise be covered by international law. If such interference occurs, the private industry would be assured only of a right to seek consultation between nations on the issue. This may be as strong a provision as space law is capable of providing. The question of whether a private industry, or a nation on its behalf, could employ forceful means to protect its activities from interference (i.e., whether it could employ self-defense) appears to be an issue which is unsettled under present space law. The Outer Space Treaty provides in Article IV that the Moon and other celestial bodies shall be used exclusively for peaceful purposes. That Article further forbids the establishment of military bases, installations, and fortifications. In addition, the proposed Moon Treaty would provide that the Moon is to be used exclusively for peaceful purposes[28] and that "[a]ny threat or use of force or any other hostile act or threat of hostile act on the moon is prohibited.[29] It is likewise prohibited to use the Moon in order to commit any such act or to engage in any such threat in relation to. . .spacecraft, the personnel of spacecraft or man-made space objects."[30]

---

[27]"Outer Space Treaty," *supra* note 1, art. III.

[28]"Moon Treaty," *supra* note 13, art. III, para. 1.

[29]*Id.*, art. III, para. 2.

[30]*Id.*

With regard to interference in its operations by others, I believe there is a substantial degree of insecurity with which the private industry would have to cope and the degree of perceived security would depend upon the degree to which such industry assesses the relief provided by international law and the ability of nations to settle claims among themselves.

E. *Stability of Legal Regime*

Private industry relies heavily upon the stability of the existing legal regime to provide security for its activities. To the extent that the legal regime is uncertain, changing and indeterminable, the insecurity of private industry in its activities will increase.

Under the existing Outer Space Treaty, a certain degree of stability exists for space activities. The right of private industry to participate in outer space activities has generally been established by space law and in practice. The general legal regime with its relative stability provides a certain level of security to private industry in its outer space activities. However, concerns over changes to the particular treaties, laws, or principles which make up that existing regime do introduce an area of instability. Private industry is concerned and, therefore, somewhat insecure about the future of communications satellites with respect to the ability to operate in the geostationary orbit on a first-come, first-serve basis as has been the practice in the past. The allocation of frequencies and the changes which may come about in that area introduce insecurity to private industry.

The proposed Moon Treaty is seen as a potentially destabilizing force for private industry participation in outer space activities. This insecurity arises from the fact that, although the proposed Moon Treaty would seem to permit private industry participation until "the exploitation of the natural resources of the moon. . .is about to become feasible,"[31] the extent and nature of private industry participation thereafter is unclear. At that point, the treaty would require that the States Parties to the agreement undertake to establish "an international regime, including appropriate procedures" to govern such exploitation.[32] Thus, the right to exploit the natural resources of the Moon and other celestial bodies is not necessarily secured for private industry. Such rights would be governed by an international regime, the composition of which is presently indeterminable. To the extent that such legal regime can be divined or determined by reference to the international regime which has been extensively negotiated and defined under the proposed Law of the Sea Treaty,[33] the security of private industry participation in outer space activities would be considerably reduced. Its right to participate, or to continue operating, or to retain its technology, or to retain and control the benefit derived from its activities would all be subjected to the economic, political, and nationalistic considerations of an international body or organization and the member states thereof, and private industry or its nation (state party) may have little or no ability to protect the interests of private industry in outer space activities.

---

[31]*Id.*, art. XI, para. 5.

[32]*Id.*

[33]For text, see Draft Convention on the Law of the Sea, U.N. Doc. A/conf. 62/L. 78, of Aug. 28, 1981.

Such insecurity alone could be sufficient to prevent the participation of private industry in outer space activities since it may preclude the industry from investing its own resources and from being able to obtain the financial backing necessary to support outer space activities. Private industry relies heavily upon private financial investment to support its activities. With the knowledge that rules will be established in the future to govern such activity and without any guidelines to allow an economic evaluation of the potential return on such activities, investment sources may hesitate or refuse to provide the financing necessary to support the entry into and continuance in outer space activities by private industry.

F. *Protection of Intellectual Property*

The intellectual property of private industry is vital to its existence. The information and technology which make up the proprietary data and trade secrets of a private industry are the lifeblood of that industry. To the extent that the right to retain and protect such technology is diluted or lost, the industry will be weakened or destroyed. Thus, a vital issue of security to private industry in its outer space activities is its ability to maintain its proprietary position.

The Outer Space Treaty does not specifically address or affect these rights. The statement in Article I that "[t]he exploration and use of outer space. . .shall be carried out for the benefit and in the interests of all countries, irrespective of their degree of economic or scientific development" could be interpreted to mean that all countries have the right to participate in such activities and, therefore, have the right to the technology involved therein. Or, this could be interpreted to the effect that all countries should share in the benefits derived therefrom. Without any provision directly addressing technology, I would not interpret the Outer Space Treaty to require private industry to relinquish its proprietary interests in technology.

The Outer Space Treaty also requires states conducting activities in outer space to inform the Secretary General of the United Nations as well as the public and the international scientific community of the "nature, conduct, locations and results of such activities."[34] I believe this requirement can be met without the necessity to reveal proprietary technology.

Article XII of the Outer Space Treaty provides that "[a]ll stations, installations, equipment and space vehicles on the moon and other celestial bodies shall be opened to representatives of other States Parties to the Treaty on the basis of reciprocity." This article does require advance notice of a projected visit[35] appropriate consultation[36] and avoidance of interference with normal operations.[37] Under such conditions, it appears likely, although not certain, that a private industry could take appropriate measures to protect is proprietary technology from disclosure during any such visit by another Party.

---

[34] "Outer Space," *supra* note 1, art. XI.

[35] *Id.*

[36] *Id.*

[37] *Id.*

Thus, I would conclude that the security of the private industry's proprietary technology or intellectual property is not substantially threatened by the provisions of the Outer Space Treaty.

The proposed Moon Treaty would include provisions substantially similar to those in the Outer Space Treaty discussed above with respect to exploration and use for the benefit of all countries, and informing the Secretary General of the United Nations, the public, and the international scientific community of space activities. The provisions allowing for inspection by one party of another party's space vehicle, equipment, facilities, stations and installations on the Moon are more detailed than in the Outer Space Treaty. To the extent that exposure of proprietary technology might be required to permit another party to assure the activities are in accordance with the law, an area of insecurity is introduced for private industry.

The proposed Moon Treaty would require that when exploitation of the natural resources of the Moon is about to become feasible, the parties must undertake to establish an international regime to govern such exploitation[38] This in itself does not address proprietary technology. However, drafts of the Law of the Sea Treaty[39] which elaborated on the concepts of "common heritage of mankind" and an international regime, have required the turning over of proprietary technology. This introduces a substantial degree of insecurity for private industry in its attempts to protect its technology. Although one cannot be sure at this time whether the Moon Treaty "international regime" would follow the pattern of the Law of the Sea Treaty, the uncertainty alone creates insecurity for private industry with regard to its future participation in outer space activities on celestial bodies.

G. *Retention of the Benefits of Outer Space Activities*

The ultimate purpose of private industry is to derive a profit from its industrial activities. Generally, without a reasonable opportunity to obtain and retain a profit from an activity, private industry has little, if any, reason to engage in such activity. Likewise, with respect to activities in outer space, the security of private industry is directly related to its ability to obtain and retain a reasonable profit from those activities.

Except to the extent that the Outer Space Treaty Article I provision that "[t]he exploration and use of outer space. . .shall be carried out for the benefit and in the interests of all countries" might be interpreted to require the sharing of the "benefits" or profits of outer space activities, I do not believe that the Outer Space Treaty really affects the ability of private industry to obtain and retain profits on its outer space activities. In practice, private industry presently engaged in outer space communications activities does control and retain its profits. Under the existing Outer Space Treaty, private industry is relatively secure with respect to this all-important issue.

However, the proposed Moon Treaty states that the natural resources of the Moon are "the common heritage of all mankind."[40] It requires the establishment of an

---

[38] "Moon Treaty," *supra* note 13, art. XI, para. 5.

[39] *See supra* note 33.

[40] "Moon Treaty," *supra* note 13, art. XI, para. 1.

international regime to govern the exploitation of resources when such exploitation is about to become feasible.[41] It states that a purpose of the international regime shall be "an equitable sharing of the benefits derived from the resources."[42] The ability of private industry to obtain, control and retain profits from outer space activities under the proposed Moon Treaty is not secure. Further, as previously alluded to, the clarification of similar provisions in the negotiation of the draft Law of the Sea Treaty[43] leads to greater insecurity in this area for private industry.

V. CONCLUSION

Under the existing general space treaties, private industry is relatively secure in its participation in outer space activities. This is borne out by the present participation, especially in outer space communications projects. However, increasing numbers of abandoned manmade space objects in orbit introduce a growing area of insecurity which would be an appropriate area to be addressed by additions to the body of space law. More binding liability provisions and dispute resolution provisions would enhance the security of private industry in outer space. To provide security with respect to the initial right to participate, the right to continue activities, the protection of proprietary technology and the ability to obtain, control and retain profits from outer space activities, the uncertainties brought about by the proposed Moon Treaty provisions relating to the common heritage of mankind and to an international regime to govern exploitation must be resolved. Either these provisions must be removed, clarified or restricted, or the projected legal regime must be definitized in the very near future. Without such action, a substantial degree of uncertainty and insecurity will continue to exist with respect to outer space activities governed by the proposed Moon Treaty. So long as such uncertainty and insecurity exist the necessary financial investment will be difficult for private industry to obtain to support the activities necessary to make exploitation of the natural resources of the Moon feasible.

---

[41] *Id.*, para. 5.

[42] *Id.*, para. 7[d].

[43] *See supra* note 33.